

WORK SHEET - 05

P-BLOCK ELEMENT

Single correct

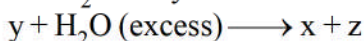
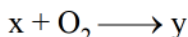
- Q.1 Cl-O bond length in ClO_2^- , Cl_2O , ClO_2^+ and ClO_2 follows the order :
 (A) $\text{ClO}_2^- < \text{Cl}_2\text{O} < \text{ClO}_2 < \text{ClO}_2^+$ (B) $\text{ClO}_2^+ < \text{ClO}_2 < \text{ClO}_2^- < \text{Cl}_2\text{O}$
 (C) $\text{ClO}_2^+ < \text{ClO}_2^- < \text{Cl}_2\text{O} < \text{ClO}_2$ (D) $\text{ClO}_2^+ < \text{Cl}_2\text{O} < \text{ClO}_2 < \text{ClO}_2^-$

- Q.2 Boron + air $\xrightarrow{\Delta}$ 'X'
 $\downarrow \text{H}_2\text{O}$
 'Y' + 'Z' $\uparrow$
 $\downarrow \text{CuSO}_4 (\text{aq})$
 'V'
 (deep blue colour)

The hybridisation of compound 'V' is :

- (A) dsp^2 (B) sp^3 (C) d^2sp^3 (D) sp^2
- Q.3 Which among the following are ionic carbide?
 (A) CaC_2 (B) Al_4C_3 (C) Be_2C (D) All of these
- Q.4 Which of the following is isoelectronic and isostructural with CO_2 ?
 (A) NO_2 (B) NO_3^- (C) NO_2^- (D) N_2O

- Q.5 In a Ostwald process
 $\text{NH}_3 + \text{O}_2 \xrightarrow[\text{Pt}]{\Delta} \text{x} + \text{H}_2\text{O}$



The incorrect option is :

- (A) 'x' shows paramagnetic behaviour (B) 'x' & 'y' are acidic oxide
 (C) 'z' can act as oxidising agent (D) 'y' & 'z' are acidic in nature
- Q.6 The temporary bleaching property is shown by
 (A) SO_2 (B) H_2O_2 (C) HOCl (D) CaOCl_2
- Q.7 POCl_3 prepared by the reaction of:
 (I) $\text{P}_4 + \text{SO}_2\text{Cl}_2 \longrightarrow$ (II) $\text{P}_4 + \text{SOCl}_2 \longrightarrow$
 (III) $\text{PCl}_5 + \text{SO}_2 \longrightarrow$ (IV) $\text{PCl}_5 + \text{C}_2\text{H}_5\text{OH} \longrightarrow$
 (A) I & III (B) II & III (C) I & IV (D) III & IV
- Q.8 Strongest acid among the following :
 (A) HClO_4 (B) HClO_3 (C) HClO_2 (D) HOCl
- Q.9 Which of the following oxides is / are not acidic in nature?
 (A) ClO_2 (B) N_2O (C) Cl_2O (D) NO_2

- Q.10 What is best reagent would you choose to convert MnO_4^{2-} to MnO_4^- ?
 (A) Cl_2 water (B) O_3 (C) (A) and (B) both (D) None of these
- Q.11 Phosphine is not obtained, when
 (A) Red phosphorous is heated with NaOH (B) White phosphorous is heated with NaOH
 (C) Ca_3P_2 reacts with water (D) Phosphorous trioxide is boiled with water
- Q.12 Which of the following is correct order of boiling point?
 (A) $\text{CCl}_4 > \text{SiCl}_4$ (B) $\text{SiCl}_4 > \text{CCl}_4$ (C) $\text{SiCl}_4 = \text{CCl}_4$ (D) None of these
- Q.13 Number of π bonds in borax is
 (A) Zero (B) One (C) Two (D) Three
- Q.14 Select the metal sulphide which is yellow in colour -
 (A) ZnS (B) Ag_2S (C) As_2S_3 (D) CoS

Comprehension

Question No. 15 to 17 (3 questions)

- (i) $\text{P} + \text{C (carbon)} + \text{Cl}_2 \longrightarrow \text{Q} + \text{CO} \uparrow$ (ii) $\text{Q} + \text{H}_2\text{O} \longrightarrow \text{R} + \text{HCl}$
 (iii) $\text{BN} + \text{H}_2\text{O} \longrightarrow \text{R} + \text{NH}_3 \uparrow$ (iv) $\text{Q} + \text{LiAlH}_4 \longrightarrow \text{S} + \text{LiCl} + \text{AlCl}_3$
 (v) $\text{S} + \text{H}_2\text{O} \longrightarrow \text{R} + \text{H}_2 \uparrow$ (vi) $\text{S} + \text{NaH} \longrightarrow \text{T}$

(P, Q, R, S & T do not represent their chemical symbols)

- Q.15 Compound Q has
I. zero dipole moment **II.** A planar trigonal structure
III. An electron deficient compound **IV.** A Lewis base
 Choose the correct code :
 (A) I, IV (B) I, III, IV (C) I, II, III (D) I, II, III, IV

- Q.16 Compound T is used as a/an
 (A) Oxidising Agent (B) Complexing Agent
 (C) Bleaching Agent (D) Reducing Agent

- Q.17 Compound S is
I. An odd- e^- compound **II.** $(2e^- - 3e^-)$ compound
III. An electron deficient compound **IV.** A sp^2 hybridized compound
 Choose the correct code :
 (A) III (B) I, III (C) II, III, IV (D) I, II, IV

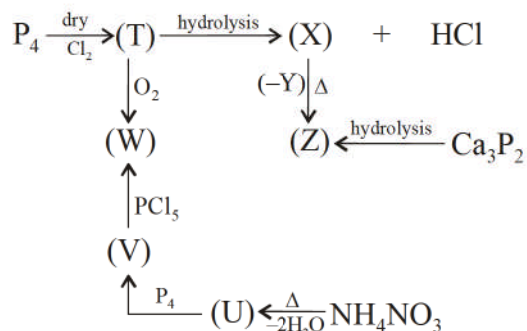
Question No. 18 to 20 (3 questions)

Hydrolysis is the reaction of water with different substance. Several mechanisms are suggested like $\text{S}_{\text{N}}1$, $\text{S}_{\text{N}}2$,..... etc.

- Q.18 The product of hydrolysis for NCl_3 is
 (A) NH_3 and HCl (B) NH_3 and HOCl (C) N(OH)_3 and HCl (D) HNO_2 and HCl
- Q.19 In which of the following cases the oxidation state of central atom gets changed on hydrolysis.
 (A) XeF_6 (B) POCl_3 (C) XeF_4 (D) HSO_3Cl
- Q.20 For which of the pair of compounds the hydrolysis products are identical.
 (A) SOCl_2 and HSO_2Cl (B) PCl_3 and AsCl_3
 (C) BiCl_3 and SbCl_3 (D) None of these



Question No. 21 to 23 (3 questions)



- Q.21 The compound (W) can also be obtained by the reaction(s).
 (A) $\text{PCl}_5 + \text{H}_2\text{SO}_4 \longrightarrow ?$ (B) $\text{PCl}_5 + \text{SO}_2 \longrightarrow ?$
 (C) $\text{PCl}_5 + \text{H}_2\text{O} \longrightarrow ?$ (D) $\text{PCl}_5 + \text{O}_2 \longrightarrow ?$
 (Equimolar ratio)
- Q.22 Select the correct statement(s)?
 (A) Heating of (X) involves disproportionation.
 (B) Heating of NH_4NO_3 to get (U) involves disproportionation
 (C) The compound (U) supports combustion.
 (D) Formation of (Z) by treating P_4 with caustic soda involves disproportionation.
- Q.23 The gas (U) is
 (A) neutral oxide
 (B) oxidising agent
 (C) also obtained by treating zinc with dilute HNO_3
 (D) linear in shape.

Question No. 24 & 25 (2 questions)

Among the oxide of nitrogen NO on oxidation changes to NO_2 which dimerizes to N_2O_4 both having different magnetic property NO by losing an electron changes to NO^+ and by gaining electron to NO^- . Similarly NO_2 changes to NO_2^+ and NO_2^- having different shapes and bond angles answer the following questions.

- Q.24 Which of the following is a neutral oxide ?
 (A) NO (B) NO_2 (C) N_2O_3 (D) N_2O_5
- Q.25 Which of the following is isoelectronic and isostructural with CO_2 .
 (A) NO_2 (B) NO_3^- (C) NO_2^- (D) N_2O

Question No. 26 & 27 (2 questions)

It is clear that $[\text{BH}_3]$ is a fugitive reaction species it exists only at exceedingly low concentration but can be isolated and studied using matrix isolation techniques. Thus it can be generated by thermal dissociation of loosely bound 1:1 ligand, H^- is a species case since it gives the symmetrical tetrahedral ion BH_4^- isoelectronic with CH_4 and NH_4^+ . Many other complexes of BH_3 with N, P As, O, S etc. donor atoms are also known and they readily formed by symmetrical homolytic cleavage of the bridge bonds in B_2H_6 .

- Q.26 $[\text{X}] + \text{Cl}_2 \longrightarrow [\text{Y}] \xrightarrow[\text{+H}_2]{\text{Electric discharge}} [\text{Z}] + \text{HCl}$
 Element Electron Deficient
 $[\text{Z}] + [\text{L}] \longrightarrow [\text{M}]$
 Adduct

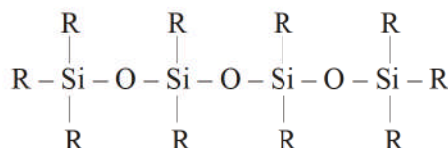
The most stable adduct formed by [Z] with [L] then L is :

- (A) PF_3 (B) CO (C) Me_2S (D) Py

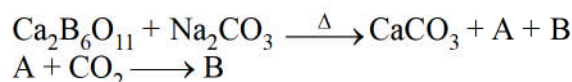
- Q.27 Choose correct option w.r.t. B_2H_6
 (A) CO gives unsymmetrical cleavage.
 (B) NH_3 , $MeNH_2$ and Me_2NH gives unsymmetrical cleavage.
 (C) H^- form less stable adduct as compare to Me_3N .
 (D) Bond length of bridge bond is less than terminal bond length in B_2H_6

More than one may be correct

- Q.28 Identify the correct statements:
 (A) Fluorine is a super halogen (B) Iodine shows basic nature
 (C) AgF is insoluble in water (D) SCN^- is a pseudohalide
- Q.29 Which of following statement is/are correct regarding graphite?
 (A) Graphite is thermodynamically more stable than diamond.
 (B) d_{C-C} among the layers and within the layers are not same.
 (C) Graphite is more reactive than diamond.
 (D) Graphite conducts electricity due to its bond polarity.
- Q.30 Choose the reactions which would liberate nitrogen gas?
 (A) $Ca(OCl)Cl + NH_3 \xrightarrow[\text{medium}]{\text{aqueous}} ?$ (B) $NH_3 + PbO \xrightarrow{\Delta} ?$
 (C) $NH_3 (\text{excess}) + Cl_2 \xrightarrow{\Delta} ?$ (D) $NH_4Cl + NaNO_2 \xrightarrow{\Delta} ?$
- Q.31 Which of the following compound(s) is / are planar about nitrogen?
 (A) $N(SiH_3)_3$ (B) $N(SiH_3)_2(BF_2)$ (C) $N(BF_2)_3$ (D) $\ddot{N}(CH_3)_3$
- Q.32 Which of the following statement(s) is/are **correct** about P_4O_6 and P_4O_{10}
 (A) Both oxides have closed cage like structure
 (B) Each oxide contains six equivalent P–O–P bonds
 (C) POP bond length in P_4O_6 is greater than P_4O_{10}
 (D) Both are the anhydrides of their respective acids.
- Q.33 In the following Silicone, select the alkyl / aryl substituted silicon chlorides which are used in formation of given Silicones:



- (A) $R SiCl_3$ (B) R_2SiCl_2 (C) R_3SiCl (D) R_4Si
- Q.34 Which of the following option is / are correct?

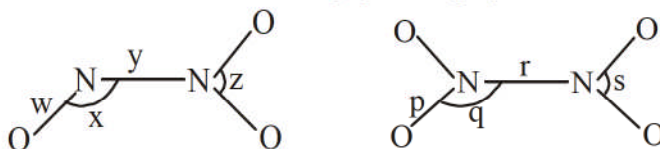


- (A) B contains 5 B–O–B linkage
 (B) A will be obtained by thermal decomposition of B
 (C) Aqueous solution of B used as a acidic buffer in titration
 (D) A exist in a polymeric form.
- Q.35 Choose the **correct** options
 (A) Conjugate base of hydrazoic acid is N_3^- .
 (B) d_{N-N} bond length in N_2H_4 is more as compare to d_{N-N} in N_2F_4
 (C) In triazine (N_3H_3) all nitrogen atoms are sp^2 hybridised
 (D) In cyclotriazine (N_3H) all N–N bond length are equal

Q.36 Choose the correct statements:

- (A) In B_2H_6 molecule bond strength of bridge bond is greater than terminal B–H bond.
 (B) In trisilyl amine maximum seven atoms are present in one plane and nitrogen has sp^3 hybridisation.
 (C) Borax give blue bead with $CuSO_4$ salt.
 (D) Borate salt on react with methanol in presence of concentrated H_2SO_4 to form a compound which give green edge flame.

Q.37 Select the **correct** orders with respect to N_2O_3 and N_2O_4 from following diagrams.



Where $\rightarrow$ solid line represents linkage :

- $\rightarrow$ w and p are NO bond length
 $\rightarrow$ x and q are ONN bond angle
 $\rightarrow$ y and r are NN bond length
 $\rightarrow$ z and s are ONO bond angle

- (A) $w < p$ (B) $x < q$ (C) $y > r$ (D) $z < s$

Match the column

Q.38

Column-I

Column-II

[Hydrolysed products at any condition]

- | | |
|--------------|---|
| (A) NF_3 | (P) One of the product is anhydride of nitrous acid |
| (B) NCl_3 | (Q) One of the product is sp^3 hybridised. |
| (C) PCl_3 | (R) One of the product is ionic with white turbidity. |
| (D) $SbCl_3$ | (S) One of the product is acidic in nature. |
| | (T) Tautomerism is observed. |

Integer Type

Q.39 Find the total number of correct order of the mentioned properties.

- | | |
|--|---|
| (i) $HOCl > HOBr > HOI$ (Oxidising nature) | (ii) $HBrO_3 > HClO_3 > HIO_3$ (Oxidising nature) |
| (iii) $HBrO_4 > HClO_4 > HIO_4$ (Oxidising nature) | (iv) $F_2 > Cl_2 > Br_2 > I_2$ (Oxidising nature) |
| (v) $HOCl < HClO_2 < HClO_3 < HClO_4$ (Thermal stability) | |
| (vi) $NH_3 > PH_3 > AsH_3 > SbH_3 > BiH_3$ (Thermal stability) | |
| (vii) $Br_2O_5 < I_2O_5$ (Thermal stability) | (viii) $NaBr > MgBr_2 > AlBr_3$ (Melting point) |
| (ix) $VO_2^+ < Cr_2O_7^{2-} < MnO_4^-$ (Oxidising power) | |
| (x) $[Ni(H_2O)_6]^{2+} < [Ni(NH_3)_6]^{2+} < [Ni(NO_2)_6]^{4-}$ (Wavelength of the light observed) | |

Q.40 3 moles of SiF_4 are reacted with water, then calculate how many total number of moles of Hexa fluoride silicic acid are formed in product.

ANSWER KEY

Q.1	B	Q.2	A	Q.3	D	Q.4	D	Q.5	B	Q.6	A	Q.7	D
Q.8	A	Q.9	B	Q.10	C	Q.11	A	Q.12	A	Q.13	A	Q.14	C
Q.15	C	Q.16	D	Q.17	A	Q.18	B	Q.19	C	Q.20	A	Q.21	ABC
Q.22	ACD	Q.23	ABCD	Q.24	A	Q.25	D	Q.26	D	Q.27	B	Q.28	ABD
Q.29	ABC	Q.30	ABCD	Q.31	ABC	Q.32	ACD	Q.33	BC	Q.34	ABCD	Q.35	AB
Q.36	ACD	Q.37	ABCD	Q.38	(A) PS (B) QS (C) QST (D) RS	Q.39	10	Q.40	2				